

7/15/25

Ana Jaramillo
Phoenix Indian Medical Center

1 Project Summary

This project involved analyzing the HVAC energy usage of select floors at the Phoenix Indian Medical Center (PIMC). The goal was to assess the energy used under the current ventilation schedules and compare them to the energy used under an optimized schedule based on occupancy. The potential energy savings from reducing unnecessary airflow during unoccupied hours were calculated based on ASHRAE 170 standards and an energy formula.

In healthcare, energy efficiency is a large concern, as they typically consume more energy, on average, than commercial buildings, with HVAC systems being one of the largest contributions. By reducing energy usage at PIMC, broader sustainability would be supported.

This analysis resulted in an average of 20.3% of energy saved per floor when following optimized ventilation schedules.

2 Project Objectives

The purpose of this project was to identify areas where energy could be saved through HVAC system manipulations, with the goal to improve energy waste. This analysis focuses on quantifying energy savings from applying an occupancy-based ventilation schedule instead of the fulltime runtime. This study used ASHRAE 170 standards to determine required air changes per hour and room-specific needs.

3 Methodology

This section covers the approach taken to evaluate the energy saving potential discussed in this project. The key aspects of the process are broken down below:

3.1 Room Type Identification

Room identification included a range of space types, such as operating rooms (ORs), pharmacy areas, waiting room areas, lounges, patient rooms, and offices. The type of room was identified by cross-checking the automated logic (WebCTRL) system and PIMC's floor plans.

3.2 Ventilation Requirements

Each of the ventilation requirements of the room were checked using ASHRAE 170 standards, focusing on the minimum required air change per hour (ACH) and whether unoccupied shutdown was allowed in the given area. The ACH was then used to estimate the

CFM of the rooms.

To estimate airflow the following equation was used:

$$CFM = \frac{ACH \times Room\ Volume}{60}$$

Where,

CFM is the estimated air flow

ACH is the minimum required air change per hour for designated room type

Note: Room volumes were approximated.

3.3 Schedule Identification

The current HVAC schedules for individual spaces were extracted from WebCTRL. To supplement this data and gather information for an optimized schedule, each floor (1st, 2nd, 3rd) was surveyed to confirm their typical occupancy hours and identify any exceptions.

3.4 Energy calculation formula

Energy consumption was calculated separated for occupied and unoccupied areas using the following two equations

Unoccupied Periods

$$kWh = \frac{1.08 \times CFM \times (Unoccupied\ Setpoint - Supply\ Temp.) \times Load\ Factor \times Hours}{3412}$$

Occupied Periods

$$kWh = \frac{1.08 \times CFM \times (Occupied\ Setpoint - Supply\ Temp.) \times Hours}{3412}$$

Where,

CFM: estimated air flow

Unoccupied Setpoint: maximum temperature setpoint available in unoccupied periods.

Occupied Setpoint: minimum temperature setpoint available in the occupied periods

Supply Temp.: supply temperature obtained from respective air handling unit (AHU) through WebCTRL

Load Factor: only considered in the unoccupied periods to account for the larger setpoint temperature range.

Hours: number of hours the room is in designated as unoccupied

Note: For unit analysis of the above formula, refer to Appendix A

4 Results

This section summarizes the estimated energy savings.

4.1 First Floor

1. **Rooms analyzed:** 32
2. **Total savings per day:** 58.5 kWh/day
3. **Percent Reduction:** 18.5%
4. **Average Occupied Time:** 18.6 hours/day

Note: some of the rooms included in the analysis were assumed to require 24/7, such as Emergency Department(ED) spaces and therefore showed no discrepancy in occupied vs unoccupied hours.

4.2 Second Floor

1. **Rooms analyzed:** 18
2. **Total savings per day:** 34.7 kWh/day
3. **Percent Reduction:** 22.3%
4. **Average Occupied Time:** 16.4 hours/day

Note: some of the rooms included in the analysis were assumed to require 24/7, therefore some showed no discrepancy in occupied vs unoccupied hours.

4.3 Third Floor

Based on the surveys to the staff, this floor was reported to be staffed 24/7, thus no adjustments were made, and was treated as a net zero energy change floor in the analysis.

5 Discussion and Recommendations

Within the constraints of the analysis, it can be concluded that waiting rooms contributed the most to the energy savings. This is likely due to the larger size and the fact that the scheduling times could be modified. Offices and lounges also showed strong savings when setpoint and schedules were adjusted.

To improve the accuracy and usefulness of this analysis, cost savings could be added as well as the incorporation of fan power and heating loads. Furthermore, before implementing any modifications, the accuracy of pressure relationships and other critical aspects need to be confirmed to work through the modified schedule. It is also important to consider that the data gathered through this analysis is based on multiple assumptions and estimations.

6 Appendix A

$$kWh = \frac{1.08 \times CFM \times (Unoccupied\ Setpoint - Supply\ Temp.) \times Load\ Factor \times Hours}{3412}$$

Considering the unit conversion of the entire formula:

Let the following terms have the units as follows:

$$[\text{Set point} - \text{Supply term}] = F^{\circ} \quad [\text{Load Factor}] = 1 \text{ (unitless)}$$

Thus,

$$kWh = ft^3/min \cdot F^{\circ} \cdot h^1$$

Since min and h are both units of time,

$$\Rightarrow ft^3 \cdot F^{\circ}$$

also,

$$kW = \frac{BTU/h}{3412} \Rightarrow kW \cdot h = \frac{BTU}{3412}$$

Thus,

$$[kWh] = [BTU]$$

Thus the units match

¹For this to realistically work, you would need a factor to help convert units, such as 1.08, which takes into account air properties and time conversions. Source: <https://www.ruppair.com/documents/white-papers/Actual%20Air%20Density%20BTU%20Calculation.pdf>